ADMIXTURE

2024-11-14

```
res_dir <- "../results/s05_admixture/"
```

This script uses ADMIXTURE @alexanderFastModelbasedEstimation2009 through the package tidypopgen @carterTidypopgenTidyPopulation2025b to estimate ancestry proportions of individuals in the global dataset.

Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt",
    reattach_to = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld"
  )
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

```
bed <- "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.bed"
gt_as_plink(mad_bor_g1000_sum, bed)
```

```
## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.bed"
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

ADMIXTURE was run on the full, cleaned dataset of 2744 individuals and 198196 SNPs.

```
adm_res <- "../data/intermediate/admixture/unsupervised/admixture_m.RDS"
if (file.exists(adm_res)) {
  qmats <-
    readRDS(adm_res)
} else {
```

```
  tictoc::tic()
  k <- c(2:10)
  qmats <- gt_admixture(mad_bor_g1000_sum,
    k,
    n_runs = 1,
    crossval = TRUE,
    seed = 1,
    n_cores = n_cores
  )
  saveRDS(qmats, "../data/intermediate/admixture/unsupervised/admixture_m.RDS")
  time <- tictoc::toc()
}
```

ADMIXTURE was run unsupervised for k values `min(range(unique(qmats$k)))` to 10.

```
# check order
testthat::expect_equal(qmats$id, mad_bor_g1000_sum$id)
testthat::expect_equal(qmats$group, mad_bor_g1000_sum$population)
```

Reorder gen_tibble

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(Region = "Other")
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(Region = case_when(
    population %in% c("Borneo", "Sumba", "Timor") ~ "Island southeast Asia",
    population %in% c("BEB", "PJL") ~ "South Asia",
    population %in% c("CHS", "KHV") ~ "East Asia",
    population %in% c("ESN", "GWD", "LWK", "MSL", "YRI") ~ "Africa",
    population %in% c(
      "Betsimisaraka", "Merina", "Diego",
      "Sakalava_Tsimihety", "Betsileo",
      "Vezo", "Mikea", "Temoro"
    ) ~ "Malagasy",
    TRUE ~ "Other"
  ))
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(Region = case_when(
    Region == "Other" ~ "Africa",
    TRUE ~ Region
  ))

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% arrange(Region, population)
```

Reorder qmats according to population order in mad_bor_g1000_sum

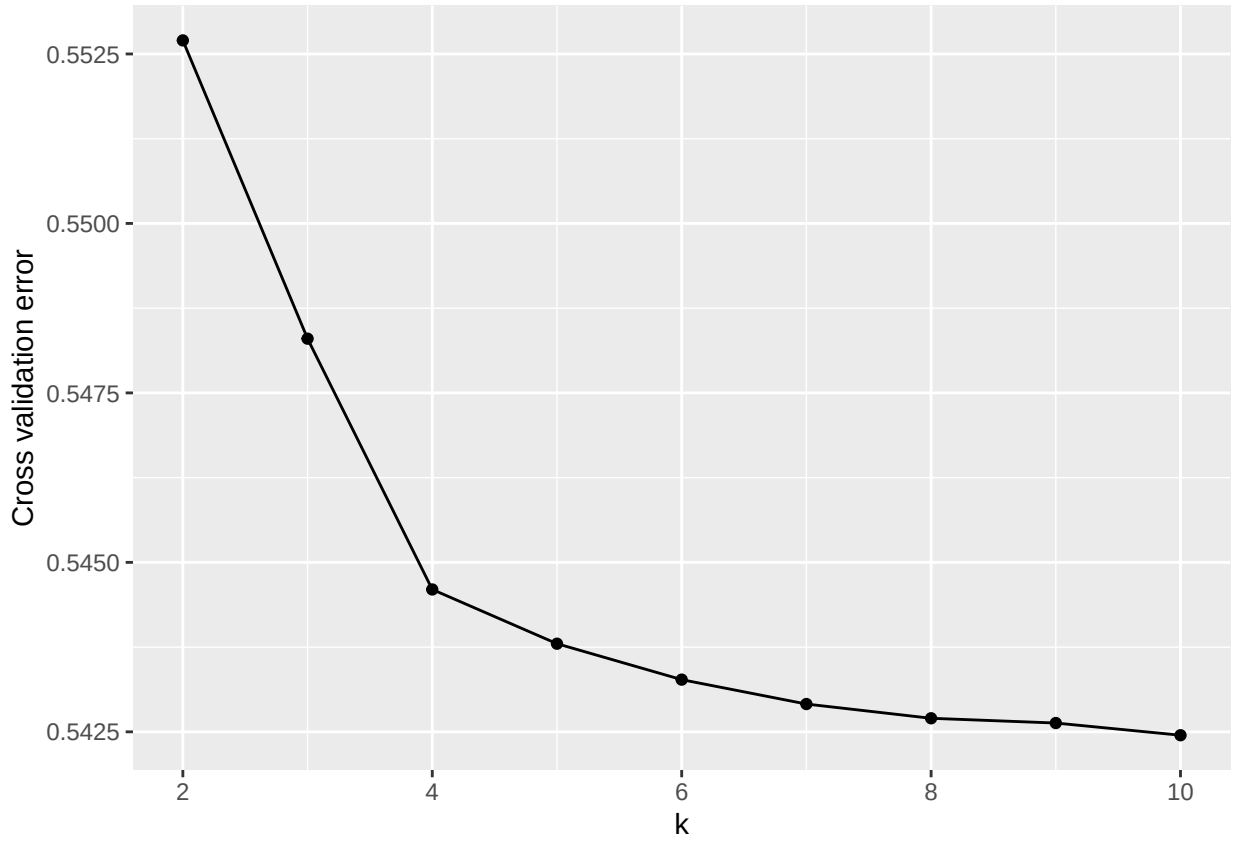
```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% filter(.data$id %in% qmats$id)
mad_bor_g1000_sum <- mad_bor_g1000_sum[match(qmats$id, mad_bor_g1000_sum$id), ]

region <- mad_bor_g1000_sum$Region
region <- factor(region, levels = unique(mad_bor_g1000_sum$Region))

qmats <- gt_admix_reorder_q(qmats, group = region)
```

CV

```
autoplot(qmats, type = "cv")
```

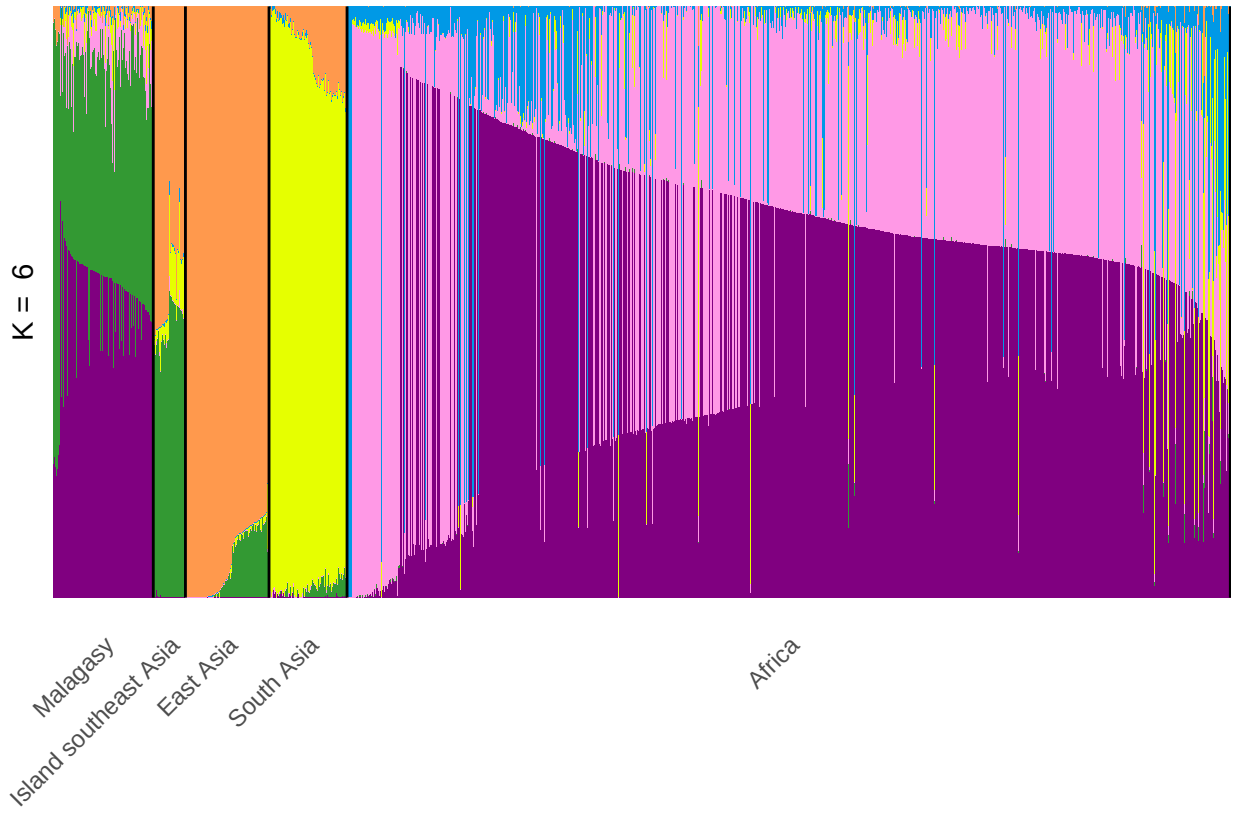



```
ggsave(paste0(res_dir, "admixture_cv_error.jpg"),
  width = 7, height = 6, dpi = 300
)
```

Cross validation error indicates that low values of K are unsuitable to describe variation in the dataset.

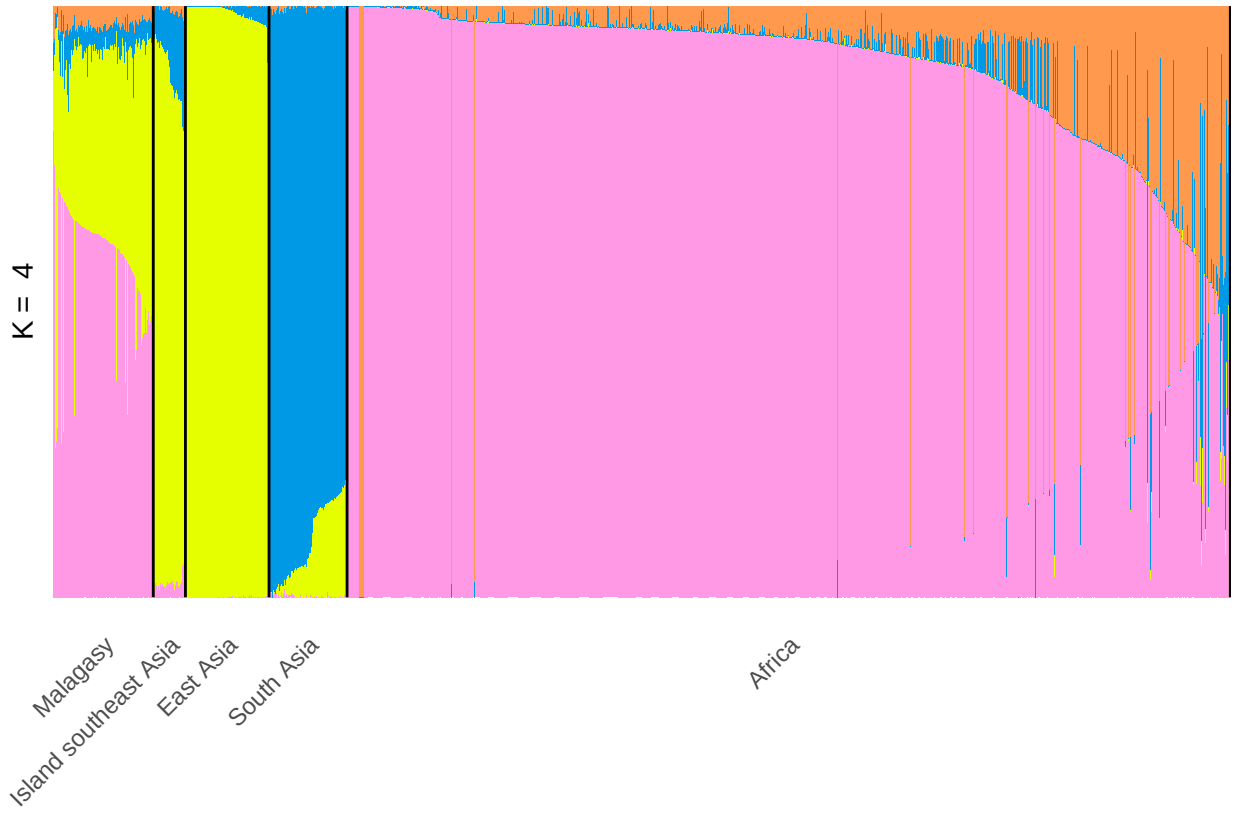
```
autoplot(get_q_matrix(qmats, k = 6, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
```

```
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
```



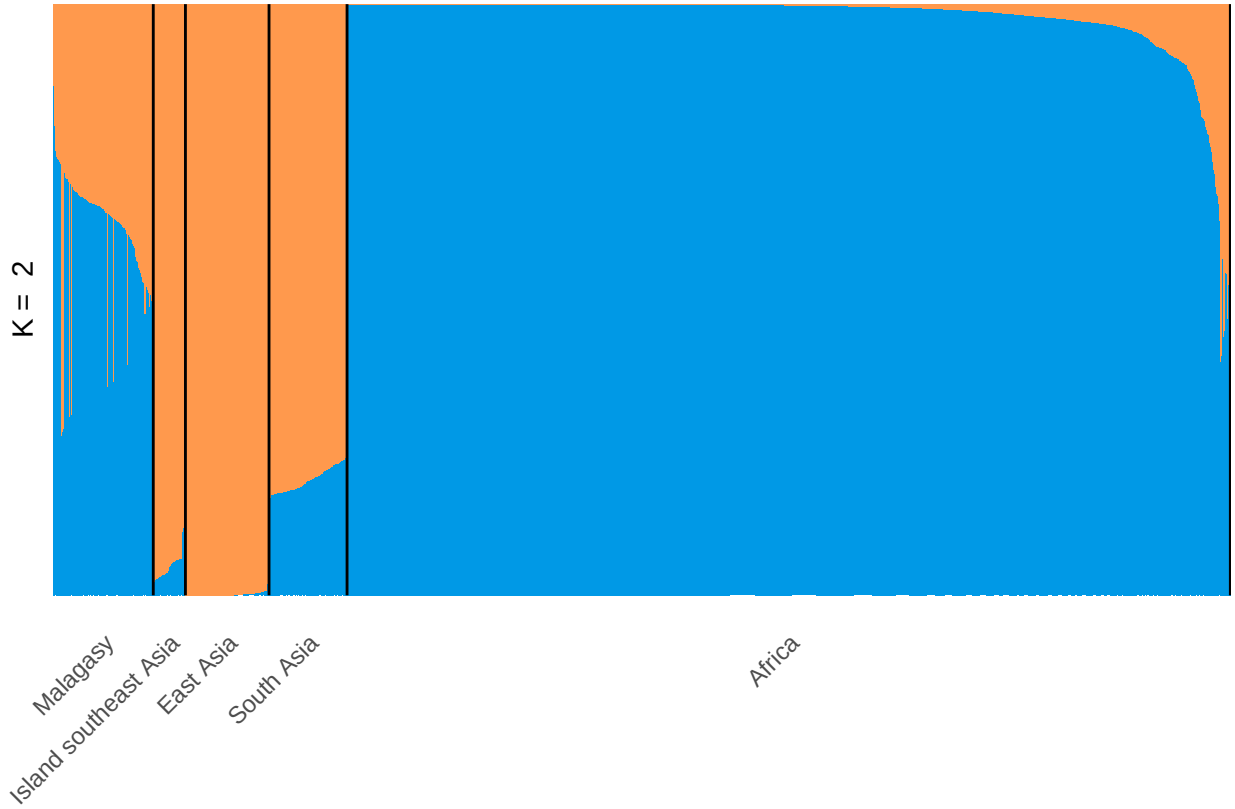

```
ggsave(paste0(res_dir, "admixture_k6.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
autoplot(get_q_matrix(qmats, k = 4, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
```




```
ggsave(paste0(res_dir, "admixture_k4.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
plot_a <- autoplot(get_q_matrix(qmats, k = 2, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
plot_a
```




```
ggsave(paste0(res_dir, "admixture_k2.jpg"),
  width = 7, height = 5, dpi = 300
)
```

Replot with labels at the top

```
group <- qmats$group
group_x <- cumsum(table(tidypopgen:::fct_inorder_base(group)))

segment_data <- data.frame(
  x = group_x + 0.5,
  xend = group_x + 0.5,
  y = rep(0, length(group_x)),
  yend = rep(1, length(group_x))
)

get_mid_points <- function(vec) {
  (vec[-length(vec)] + vec[-1L]) / 2.
}

q_tbl <- as.data.frame(get_q_matrix(qmats, k = 2, run = 1))
q_tbl$id <- attr(get_q_matrix(qmats, k = 2, run = 1), "id")
q_tbl$group <-
```

```
  rep(
    attr(get_q_matrix(qmats, k = 2, run = 1), "group"),
    each = nrow(q_tbl) / length(attr(
      get_q_matrix(qmats, k = 2, run = 1),
      "group"
    ))
  )
q_tbl$id <- seq_len(nrow(q_tbl))
q_tbl <- q_tbl %>%
  dplyr::arrange(.data$group, .data$id)
q_tbl$id <- seq_len(nrow(q_tbl))
q_tbl <- q_tbl %>%
  tidyr::pivot_longer(
    cols = dplyr::starts_with(".Q"),
    names_to = "q",
    values_to = "percentage"
  ) %>%
  dplyr::mutate(percentage = as.numeric(.data$percentage))
q_tbl <- q_tbl %>%
  dplyr::group_by(.data$group, .data$id) %>%
  dplyr::arrange(.data$group) %>%
  dplyr::mutate(
    q = factor(
      .data$q,
      levels = .data$q[order(.data$percentage, decreasing = FALSE)]
    )
  )
dominant_q <- q_tbl %>%
  dplyr::group_by(.data$id) %>%
  dplyr::summarize(dominant_q = max(.data$percentage), .groups = "drop")

q_tbl <- q_tbl %>%
  dplyr::left_join(dominant_q, by = "id")

q_tbl <- q_tbl %>%
  dplyr::group_by(.data$group) %>%
  dplyr::arrange(desc(.data$dominant_q), .by_group = TRUE)

levels_q <- unique(q_tbl$id)
q_tbl <- q_tbl %>%
  dplyr::mutate(id = factor(.data$id, levels = levels_q))


plot_a <- plot_a +
  ggplot2::geom_segment(
    data = segment_data,
    ggplot2::aes(
      x = .data$x,
      y = .data$y,
      xend = .data$xend,
      yend = .data$yend
    ),
    inherit.aes = FALSE
```

```
  ) +
  ggplot2::scale_x_discrete(
    breaks = unique(q_tbl$id)[get_mid_points(c(0, group_x))],
    labels = names(group_x),
    position = "top"
  ) +
  ggplot2::theme(
    axis.text.x = ggplot2::element_text(angle = 45, vjust = 1, hjust = 0)
  ) +
  theme(
    axis.text.x.top = element_text(size = 10),
    axis.text.x.bottom = element_blank()
  )
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```

Add Malagasy population names

```
mad_bor_g1000_sum$Region <- ifelse(mad_bor_g1000_sum$Region == "Malagasy",
  mad_bor_g1000_sum$population,
  mad_bor_g1000_sum$Region
)

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% arrange(Region, population)
```

Reorder qmats according to population order in mad_bor_g1000_sum

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% filter(.data$id %in% qmats$id)

mad_bor_g1000_sum <- mad_bor_g1000_sum[match(qmats$id, mad_bor_g1000_sum$id), ]

# rename "Sakalava Tsimihety" and "Diego"
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(Region = case_when(
    Region == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
    Region == "Diego" ~ "Diego Mixed",
    TRUE ~ Region
  ))

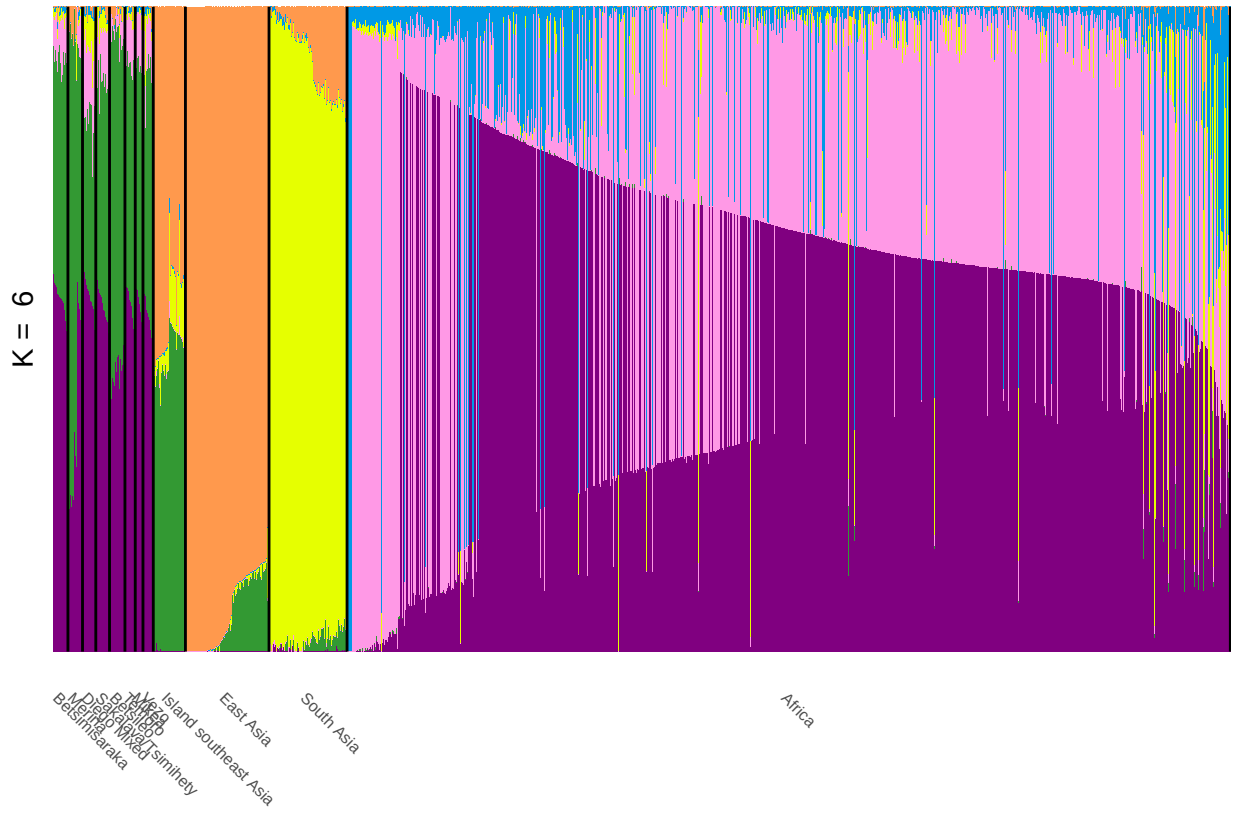
region <- mad_bor_g1000_sum$Region
region <- factor(region, levels = unique(mad_bor_g1000_sum$Region))

qmats <- gt_admix_reorder_q(qmats, group = region)
```

Plot

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  group_by(Region)

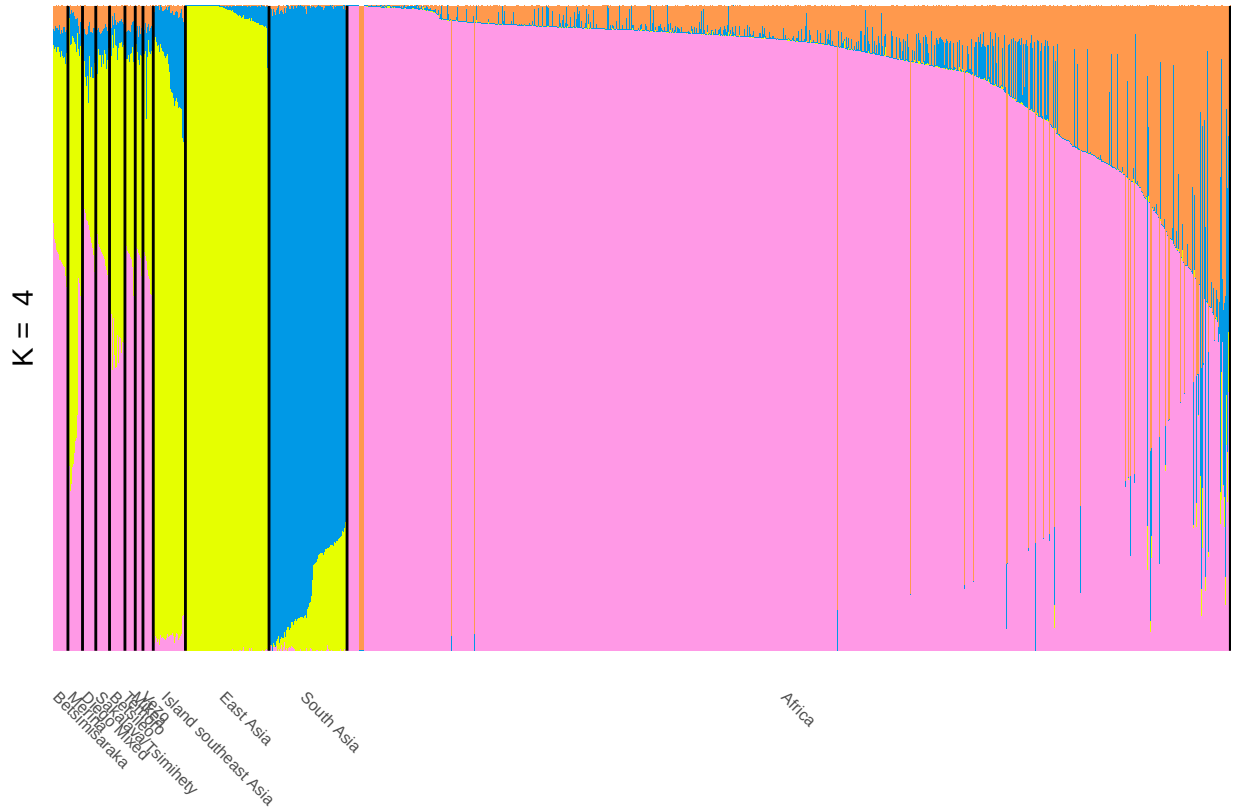
autoplot(get_q_matrix(qmats, k = 6, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
) +
  theme( # Rotate and adjust alignment
    axis.text.x = element_text(angle = -45, hjust = 0, vjust = 1, size = 6)
  )
```




```
ggsave(paste0(res_dir, "admixture_k6_malagasy_named.jpg"),
  width = 7, height = 5, dpi = 300
)

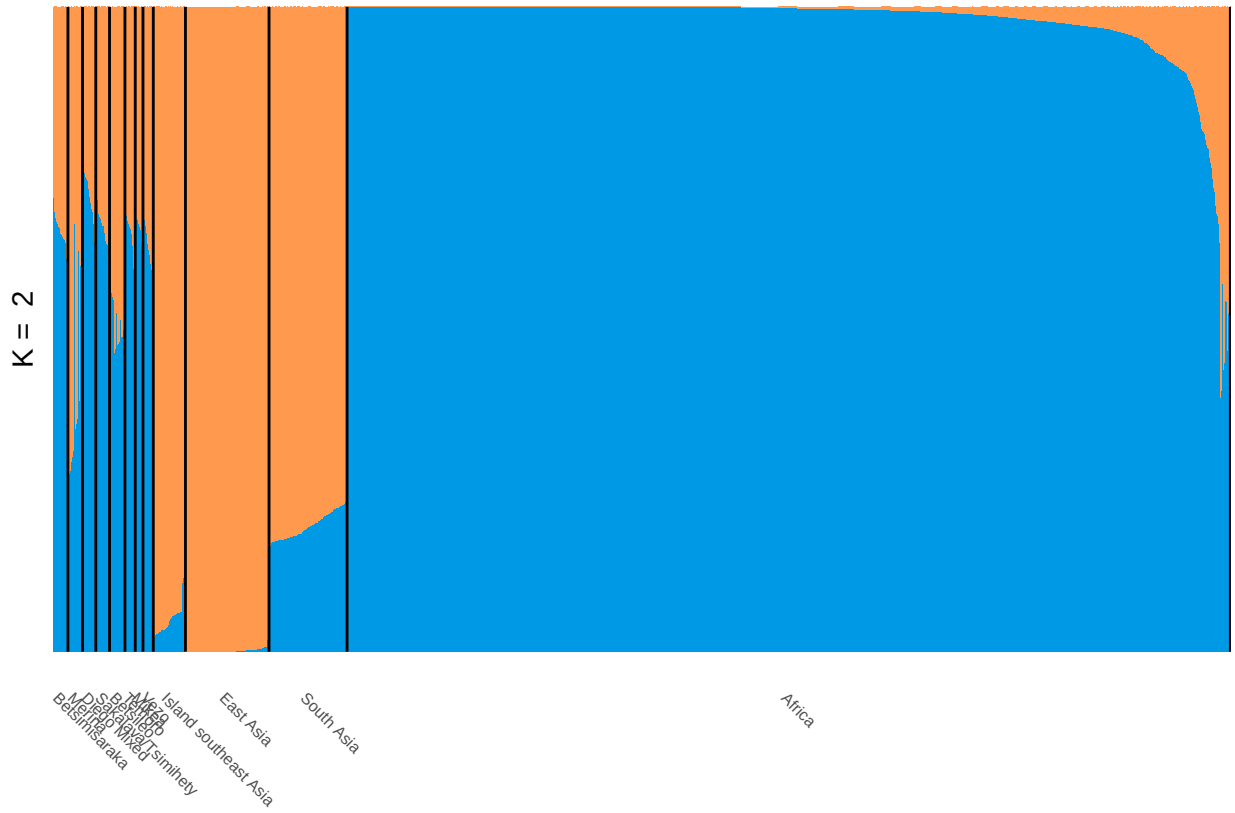

autoplot(get_q_matrix(qmats, k = 4, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
```

```
  reorder_within_groups = TRUE
) +
  theme( # Rotate and adjust alignment
    axis.text.x = element_text(angle = -45, hjust = 0, vjust = 1, size = 6)
  )
```




```
ggsave(paste0(res_dir, "admixture_k4_malagasy_named.jpg"),
  width = 7, height = 5, dpi = 300
)

autoplot(get_q_matrix(qmats, k = 2, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
) +
  theme( # Rotate and adjust alignment
    axis.text.x = element_text(angle = -45, hjust = 0, vjust = 1, size = 6)
  )
```




```
ggsave(paste0(res_dir, "admixture_k2_malagasy_named.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
tidyq <- tidy(get_q_matrix(qmats, k = 2, run = 1), mad_bor_g1000_sum)
# Min and max percentage of .Q2 in group 2 (Malagasy)
tidyq %>%
  filter(group == 2) %>%
  filter(q == ".Q2") %>%
  reframe(range = range(percentage))
```

```
## # A tibble: 2 x 1
##   range
##   <dbl>
## 1 0.319
## 2 0.740
```

```
# find samples of interest within gt_admix object `qmats`
malagasy_index <- which(qmats$group %in% c(
  "Betsimisaraka", "Merina", "Diego Mixed",
  "Sakalava/Tsimihety", "Betsileo",
  "Vezo", "Mikea", "Temoro"
))

# subset gt_admix object `qmats` to only include samples of interest
```

```r
gt_admix_subset <- function(qmats, index) {
  qmats_subset <- qmats
  # k doesn't change
  # CV doesn't change
  # loglik doesn't change
  # algorithm doesn't change
  # attributes don't change

  # subset id
  qmats_subset$id <- qmats_subset$id[index]
  # subset group
  qmats_subset$group <- qmats_subset$group[index]
  # reset the levels of group
  qmats_subset$group <- factor(qmats_subset$group,
    levels = unique(qmats_subset$group)
  )

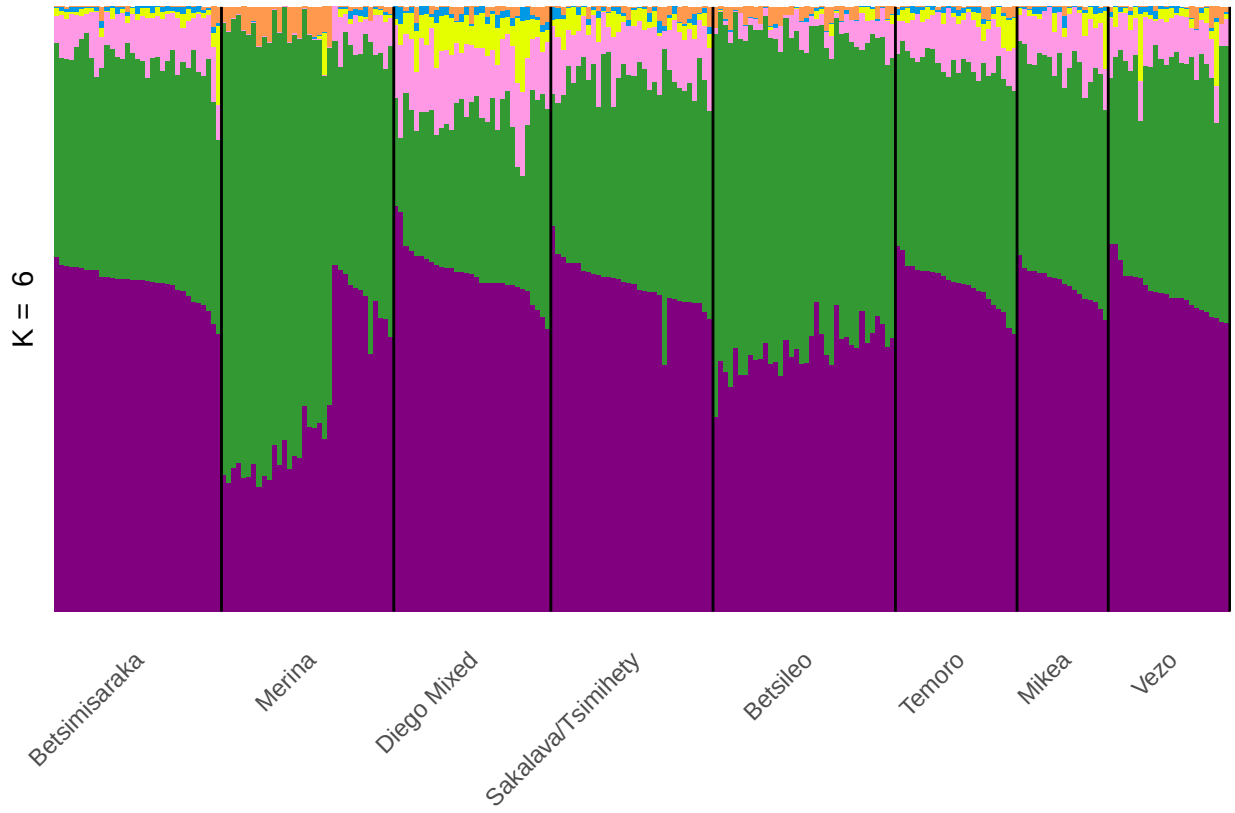
  # subset Q
  for (i in seq_along(qmats$k)) {
    qmats_subset$Q[[i]] <- qmats$Q[[i]][index, ]
    # add an id attribute
    attr(qmats_subset$Q[[i]], "id") <- qmats_subset$id
    # add a group attribute
    attr(qmats_subset$Q[[i]], "group") <- qmats_subset$group
  }

  # subset P
  for (i in seq_along(qmats$k)) {
    qmats_subset$P[[i]] <- qmats$P[[i]]
    # add an id attribute
    attr(qmats_subset$P[[i]], "id") <- qmats_subset$id
    # add a group attribute
    attr(qmats_subset$P[[i]], "group") <- qmats_subset$group
  }

  # subset each matrix
  return(qmats_subset)
}

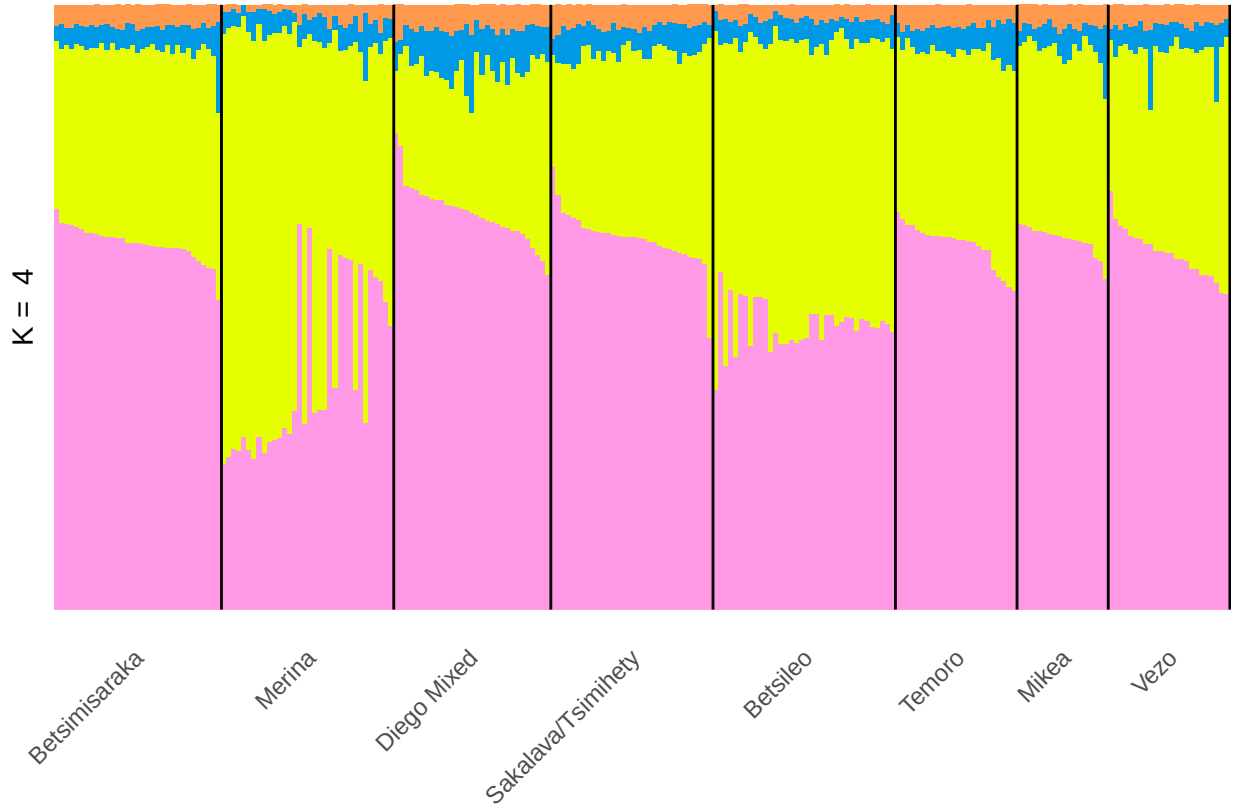
qmats_malagasy <- gt_admix_subset(qmats, malagasy_index)

autoplot(get_q_matrix(qmats_malagasy, k = 6, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
```



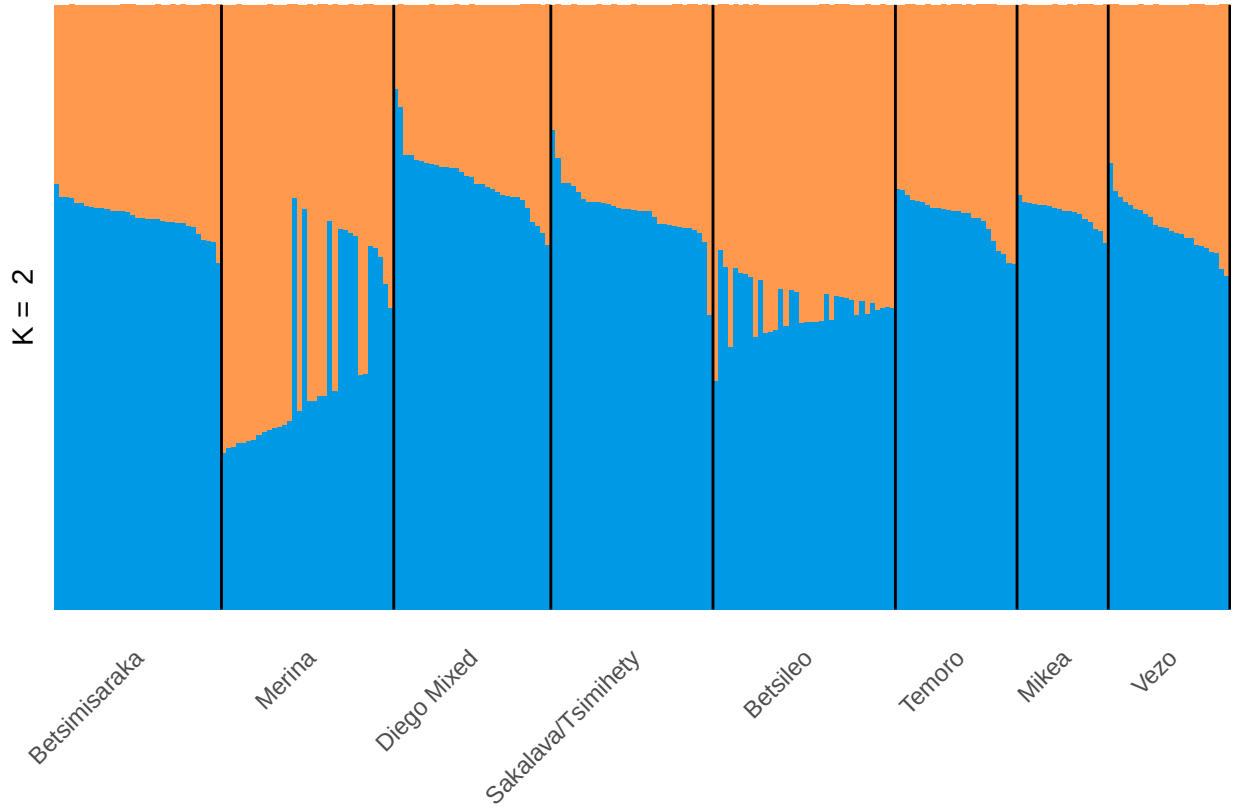

```
ggsave(paste0(res_dir, "admixture_k6_malagasy.jpg"),
  width = 7, height = 5, dpi = 300
)

autoplot(get_q_matrix(qmats_malagasy, k = 4, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
```



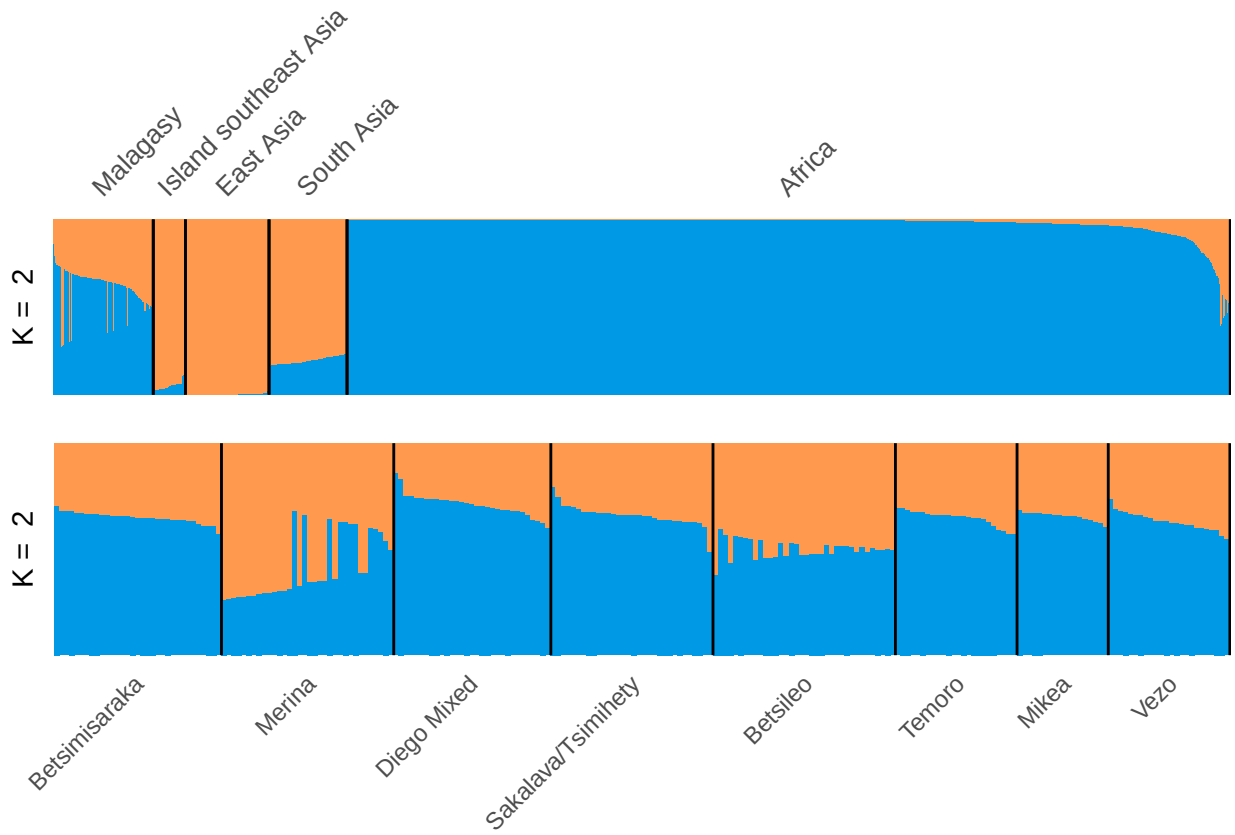

```
ggsave(paste0(res_dir, "admixture_k4_malagasy.jpg"),
  width = 7, height = 5, dpi = 300
)


plot_b <- autoplot(get_q_matrix(qmats_malagasy, k = 2, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
plot_b
```




```
ggsave(paste0(res_dir, "admixture_k2_malagasy.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
grid.arrange(plot_a, plot_b, nrow = 2)
```




```
g <- arrangeGrob(plot_a, plot_b, nrow = 2)

ggsave(g,
  filename = paste0(res_dir, "admixture_k2_inset.jpg"),
  width = 7, height = 6, dpi = 300
)

pdf(paste0(res_dir, "admixture_k2_inset.pdf"),
    width = 7, height = 6, useDingbats = FALSE)
g
```

```
## TableGrob (2 x 1) "arrange": 2 grobs
##   z     cells    name           grob
## 1 1 (1-1,1-1) arrange gtable[layout]
## 2 2 (2-2,1-1) arrange gtable[layout]
```

```
dev.off()
```

```
## pdf
##   2
```

Check which ancestry components each .Q represents

```
augmented <- augment(get_q_matrix(qmats, k = 2, run = 1),
  data = mad_bor_g1000_sum
)

augmented_chs <- augmented %>% filter(population == "CHS")
head(augmented_chs)
```

```
## # A tibble: 6 x 9
## # Groups:   Region [1]
##   id      population sex    genotypes Region    .rownames     .Q1    .Q2 group
##   <chr>   <chr>      <chr> <vctr_SNP> <chr>         <int>   <dbl> <dbl> <int>
## 1 HG00403 CHS        <NA>   [1,1,...] East Asia       308 0.00001 1.000    10
## 2 HG00404 CHS        <NA>   [1,0,...] East Asia       309 0.00001 1.000    10
## 3 HG00406 CHS        <NA>   [2,1,...] East Asia       310 0.00001 1.000    10
## 4 HG00407 CHS        <NA>   [2,1,...] East Asia       311 0.00001 1.000    10
## 5 HG00409 CHS        <NA>   [1,0,...] East Asia       312 0.00001 1.000    10
## 6 HG00410 CHS        <NA>   [1,2,...] East Asia       313 0.00001 1.000    10
```

```
augmented_gwd <- augmented %>% filter(population == "GWD")
head(augmented_gwd)
```

```
## # A tibble: 6 x 9
## # Groups:   Region [1]
##   id      population sex    genotypes Region .rownames   .Q1    .Q2 group
##   <chr>   <chr>      <chr> <vctr_SNP> <chr>      <int> <dbl>  <dbl> <int>
## 1 HG02461 GWD        <NA>   [0,0,...] Africa       685 0.980 0.0200    12
## 2 HG02462 GWD        <NA>   [2,1,...] Africa       686 0.960 0.0397    12
## 3 HG02464 GWD        <NA>   [2,0,...] Africa       687 0.992 0.0085    12
## 4 HG02465 GWD        <NA>   [1,0,...] Africa       688 0.984 0.0156    12
## 5 HG02561 GWD        <NA>   [1,0,...] Africa       689 0.967 0.0330    12
## 6 HG02562 GWD        <NA>   [0,0,...] Africa       690 0.981 0.0192    12
```